Hybrid Modeling of Wildfire Spread in Southern California

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Context

In Spring 2026, I attended a hackathon at UCSD, in collaboration with the CalFire department. The aim was to make solutions to prevent/curb disasters caused by SoCal wildfires. I got a chance to talk to the CalFire department regarding the struggles they encounter in their workflows, what causes unforeseen fires, why things can go out of control, and what kind of tech they currently use to deal with such situations. Motivated by this experience, I decided to dedicate my capstone project to improve the current fire-spread prediction methods with my newly gained knowledge, so I reviewed research literature on fire-spread models (the current benchmark is the Rothermel model), and came up with this proposal.

Introduction

Southern California's most destructive fires, such as Thomas (2017) and Woolsey (2018), were driven by Santa Ana winds, when embers start spot fires far ahead of the main front. Rothermel-based spread models are interpretable but biased and treat spotting as an add-on; data-driven models learn from satellite observations but generalize poorly to unseen fires. This project builds a hybrid: a Rothermel core, an ember-spotting module, and a learned correction of the core's errors.

Prior Work

Rothermel (1972) underpins U.S. operational fire behavior systems (Andrews, 2018); FARSITE (Finney, 1998) propagates it across landscapes with spotting from Albini's (1979) lofting model, and Koo et al. (2010) review firebrand transport and ignition. Chen et al. (2022) tracked California fire growth at 12-hour steps from VIIRS, and the Next Day Wildfire Spread (Huot et al., 2022) and WildfireSpreadTS (Gerard et al., 2023) benchmarks train deep networks on next-day fire masks. Rochoux et al. (2014) corrected spread model parameters by assimilating observed fronts. This project joins these strands and evaluates on fires withheld from training.

Dataset

Target: Chen et al. (2022; figshare; GeoPackage/NetCDF/CSV): 12-hourly VIIRS perimeters and fire lines for California 2012–2020 at 375 m, including 735 large fires (> 4 km²) with 12,801 half-daily growth records. The Southern California subset is the study set, split by fire to avoid spatial leakage. **Predictors:** LANDFIRE fuel models and canopy (30 m GeoTIFF), USGS 3DEP slope and aspect, and HRRR hourly wind, humidity, and temperature (3 km, GRIB2) checked against RAWS. Each sample is one fire step: current perimeter plus ~10–15 raster features in, next perimeter out. CAL FIRE FRAP perimeters give an independent check; WildfireSpreadTS (607 U.S. fires, 23 channels) can pretrain the ML baseline.

Intended Tasks and Milestones

1. **Problem Definition**

The question is whether combining a Rothermel spread model, an ember-spotting module, and a learned residual correction predict 12-hour fire perimeters better than physics-only or ML-only models on held-out Southern California fires. **H1:** The hybrid beats both baselines (IoU, Sørensen index). **H2:** Spotting gives its largest gains in Santa Ana wind periods. Scope: Select the study fires and fix the evaluation metrics (IoU, Sørensen index, area error).

2. **Data Collection**

Target data: the Chen et al. (2022) California VIIRS fire-spread dataset, which has 12-hourly perimeters for 2012–2020. Input layers: LANDFIRE fuels and canopy, USGS 3DEP terrain, and HRRR weather, checked against RAWS stations. Cross-check: CAL FIRE FRAP perimeters.

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3. Data prep

Reproject everything to one UTM grid and align the layers. Build one sample per fire per time step, with the current perimeter and features as input and the next perimeter as the label. Split train, validation and test sets by fire rather than by pixel, to avoid spatial leakage.

4. Exploratory Data Analysis

Examine how spread rate varies with wind, slope and fuel type. Compare Santa Ana and non-Santa Ana periods. Look for signs of spotting, such as burned areas that appear ahead of the main front. This gives early evidence for my hypothesis that spotting matters most in high-wind conditions.

5. Modeling

Build four models:

- a Rothermel physics baseline
- the spotting module (fire intensity, ember lofting, landing distribution, ignition probability)
- an ML-only baseline (U-Net)
- the hybrid model, which uses ML to correct the physics output

6. Evaluation and Deployment

Evaluate all models on held-out fires using the metrics above, split by wind regime, and ablate each component. Deliver the code repository and an interactive map demo showing predicted versus observed perimeters. Write the final report.